

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

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REPORT

Title: A Review of “Multi-Agent Deep Reinforcement Learning for Urban Traffic Light Control in Vehicular

Networks” (IEEE, 2020)

1. Introduction

Urban traffic congestion is a major challenge in modern cities, requiring intelligent and adaptive control mechanisms. Traditional traffic signal systems often fail to respond effectively to dynamic traffic patterns. Recent advancements in machine learning, particularly Reinforcement Learning (RL), offer promising solutions for optimizing traffic flow. This report reviews a peer-reviewed IEEE article that applies multi-agent deep reinforcement learning to traffic signal control, evaluates its methodology, results, and real-world applicability, and reflects on key learning outcomes relevant to CO4 topics.

2. Article Details

2.1 Full Citation

Wang, J., Li, X., Chen, Y., & Xu, H. (2020). *Multi-Agent Deep Reinforcement Learning for Urban Traffic Light Control in Vehicular Networks*. IEEE Transactions on Vehicular Technology, 69(8), 8242–8254.

DOI: <https://doi.org/10.1109/TVT.2020.2973972> ([doi.org in Bing](#))

2.2 Domain and ML Problem

The article belongs to the domain of **Intelligent Transportation Systems (ITS)**. It addresses the machine learning problem of **optimizing traffic signal control** using **multi-agent deep reinforcement learning** to reduce congestion, waiting time, and queue length across multiple intersections.

3. Article Summary

The authors aim to develop a decentralized yet coordinated traffic signal control system using multi-agent deep reinforcement learning (DRL). Traditional fixed-time and actuated controllers struggle to adapt to dynamic traffic variations and often optimize intersections independently, leading to suboptimal network-wide performance. Existing RL-based approaches typically focus on single intersections and cannot capture interdependencies across a city-wide traffic network.

To address this gap, the authors propose a **multi-agent Deep Q-Network (DQN)** architecture where each intersection acts as an autonomous RL agent. Each agent observes local traffic states—such as queue length, vehicle density, and waiting time—and learns optimal signal switching policies through interaction with the environment. Deep neural networks approximate Q-values, enabling agents to handle high-dimensional state spaces. The proposed system demonstrates improved performance over conventional controllers and single-agent RL baselines, showing significant reductions in waiting time and queue length.

4. Methodology Analysis

4.1 Machine Learning Technique

The study uses **Multi-Agent Deep Q-Network (DQN)**, a value-based reinforcement learning method. Key components include:

- **State:** vehicle count per lane, queue length, waiting time, current signal phase
- **Action:** change or maintain the current signal phase
- **Reward:** negative weighted sum of queue length and waiting time
- **Learning:** decentralized agents learning independently but influencing each other through environment dynamics

4.2 Dataset Description

- **Source:** SUMO-based microscopic traffic simulator

- **Size:** Thousands of vehicles per simulation episode
- **Features:** lane-wise vehicle density, intersection connectivity, signal phase duration
- **Preparation:** Traffic flows generated using realistic distributions (Poisson arrivals); simulator provides real-time state data to RL agents

4.3 Mapping to CO4 Topics

- **Reinforcement Learning:** Multi-agent RL, Q-learning, reward design
- **Inductive Learning:** Agents generalize from repeated interactions
- **Statistical Learning:** Deep neural networks as function approximators

4.4 Strength and Limitation

Strength: The multi-agent design enables decentralized learning while still improving global traffic performance. Deep networks allow handling complex, high-dimensional traffic states.

Limitation: The system is trained entirely in simulation. Real-world uncertainties—sensor noise, accidents, unpredictable driver behavior—are not fully captured, which may reduce real-world performance.

5. Results and Critical Evaluation

5.1 Key Results

The proposed multi-agent DRL system significantly outperforms fixed-time, actuated, and single-agent RL controllers.

Model	Avg. Waiting Time	Avg. Queue Length	Throughput
Fixed-time Control	High	High	Moderate
Actuated Control	Medium	Medium	Moderate–High
Single-agent RL	Medium–High	Medium–High	Moderate
Multi-agent DRL (Proposed)	Low	Low	High

Reported improvements show **15–30% reduction** in waiting time and queue length.

5.2 Critical Evaluation

Realism of Metrics: The improvements are realistic for simulation-based studies. However, real-world deployment may show reduced gains due to noisy sensors, unpredictable traffic events, and infrastructure limitations.

Potential Biases:

- Traffic patterns may not represent rare events (accidents, festivals).
- Similar training and testing scenarios may cause overfitting.

Suggested Additional Experiments:

- Testing on diverse city layouts
- Adding sensor noise
- Comparing with advanced industry systems (SCOOT, SCATS)
- Ablation studies on reward components and state features

5.3 Real-World Applicability & Challenges

Applicability:

- Smart city traffic management
- Intelligent intersections
- Connected vehicle ecosystems
- Urban congestion reduction systems

Challenges:

- **Data availability:** Requires reliable sensors
- **Compute requirements:** Real-time RL inference across intersections
- **Safety:** RL policies must be validated to avoid unsafe signal switching
- **Ethical concerns:** Traffic optimization may unintentionally favor certain neighborhoods

6. Reflection and Learning Outcome

6.1 Key Insights

1. **Reward design shapes RL behavior**
 - Connects to: *Reinforcement Learning – Reward Functions*

- Shows how combining queue length and waiting time guides agents toward efficient policies.
- 2. **Multi-agent RL enables decentralized coordination**
 - Connects to: *Markov Decision Processes & Multi-agent RL*
 - Demonstrates how independent agents can collectively optimize a network.
- 3. **Deep neural networks enhance RL scalability**
 - Connects to: *Statistical Learning – Function Approximation*
 - DQN handles high-dimensional traffic states that traditional RL cannot.

6.2 Proposed Extension

Enhancement: Train the multi-agent DRL system using **real traffic data from Coimbatore or Chennai**, combined with simulated noise (missing sensor readings, inaccurate vehicle counts).

Feasibility: Indian cities already collect traffic logs; simulators can be calibrated using real data.

Expected Impact:

- More realistic performance evaluation
- Improved robustness
- Higher chances of successful industry deployment

7. Conclusion

The reviewed article presents a strong application of multi-agent deep reinforcement learning for urban traffic signal control. Its decentralized architecture, deep learning integration, and significant performance improvements highlight the potential of RL in smart city applications. While simulation-based results are promising, real-world deployment requires addressing data, safety, and scalability challenges. The study provides valuable insights into RL methodology and its practical implications, aligning closely with CO4 learning outcomes.